

Gleb Arestov

Lead Frontend / Browser Rendering Engineer

Batumi, Georgia (UTC+4) | Open to remote work, relocation for stronger timezone overlap, and short onboarding trips
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SUMMARY

Principal frontend/browser-runtime engineer with 15+ years of experience building high-performance browser-based, desktop, video, visualization, and local-first applications. Strong in TypeScript, React, Electron, Chrome Extensions, browser rendering, SVG/canvas visualization, WebCodecs, custom state runtimes, performance profiling, memory analysis, browser automation, and configuration-driven UI systems. First engineer at Sidekick Browser, a browser startup later acquired by Perplexity.

CORE SKILLS

Frontend: TypeScript, JavaScript, React, Redux, Svelte, HTML/CSS, SVG, Canvas, OffscreenCanvas, Lottie, WebGPU, WebCodecs.
Browser/Desktop: Electron, Chrome Extensions, Chrome DevTools Protocol, BrowserView, IPC, browser automation, cross-context coordination.
Runtime/State: DKT, graph-based state, derived state, dependency tracking, incremental recomputation, hydration, persistence, local-first/offline-first.
Video/Rendering/Testing: browser rendering, animation systems, WebRTC, CRDT, Playwright, Puppeteer, WebDriver, FFprobe, screenshot/rendering tests, CI/CD.
Performance: rendering optimization, memory profiling, heap snapshots, Chrome DevTools, Workers, GC pressure reduction, object reuse/pooling.
Backend/Storage: Node.js, PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, LevelDB, IndexedDB, AWS, Docker, Clojure, Python.

SELECTED RENDERING / VIDEO / VISUALIZATION WORK

Minicut Video Editor - Local-first collaborative browser video editor with browser rendering, WebCodecs workflows, CRDT collaboration, WebRTC sync, Playwright automation, and FFprobe-based rendering tests.

<https://minicut-video-editor.pages.dev/>

Weather App - State-replicated weather app with animated icon rendering via Worker, OffscreenCanvas, and Lottie.

<https://weather-app-root.pages.dev/>

Datalaboratory / Moscow Marathon Visualization - Performance-sensitive JavaScript/SVG visualization implemented with vanilla JS DOM/SVG rendering, lower-level than D3, for tighter control over animation and performance.

<https://arestov.github.io/linkkraft-notes/cv-media/Moscow%20Marathon%20Visualization.mp4>

Motion Tailor - Cloud platform for customized video production and rendering, with React/Redux UI, Node/PostgreSQL backend pieces, browser QA automation, screenshots, Docker, and CI.

<https://arestov.github.io/linkkraft-notes/cv-media/MotionTailor%20UI.mp4>

Linkkraft - Offline-first Electron browser with DOM snapshotting, deterministic visual replay, canvas-based history-tree visualization, and custom runtime/state architecture.

<https://arestov.github.io/linkkraft-notes/trails-tree-plus-offline-for-tool-for-thoughts/linkkraft-fresh-offline-spa.mp4>

EXPERIENCE

Self-Employed / Independent R&D

Principal Engineer / Product Prototyper | Mar 2026 - Present

- Building browser/runtime and local-first prototypes around DKT, React, Svelte, WebRTC, CRDT, WebCodecs, WebGPU, Electron, Playwright, and AI-assisted workflows.
- Exploring state replication, collaborative editing, graph-based runtime visualization, AI workflow orchestration, experimental browser UI/rendering models, and browser-based video workflows.

Selected prototypes:

- **Minicut Video Editor:** local-first collaborative video editor with browser rendering, WebCodecs, CRDT, WebRTC, Playwright, and FFprobe-based rendering tests. <https://minicut-video-editor.pages.dev/>
- **Weather App:** state replication across tabs/devices; animated weather icons via Worker, OffscreenCanvas, and Lottie. <https://weather-app-root.pages.dev/>
- **DKT Meta Graph Visualization:** interactive visualization of code/runtime relationships with Svelte and XYFlow. <https://arestov.github.io/linkkraft-notes/cv-media/linkkraft-dkt-meta-graph-xyflow.mp4>
- **Canvas Browser Prototype:** experimental browser UI/rendering prototype with Electron, WebGPU, and shared texture. <https://arestov.github.io/linkkraft-notes/cv-media/canvas-browser-prototype.mp4>
- **AI Orchestration Prototype:** desktop prototype for coordinating AI-assisted workflows with DKT, Electron, and Svelte. <https://arestov.github.io/linkkraft-notes/cv-media/ambition-prototype.mp4>

Career Break / Non-IT Work & Personal Programming

Dec 2023 - Feb 2026

- Worked outside full-time software engineering while continuing personal programming, technical reading, and experimentation with browser/runtime and local-first ideas.
- Maintained hands-on interest in DKT-style state models, frontend architecture, browser-based applications, and high-performance interactive UI.

ATAD ML

Contract Engineer | Nov 2023

- Built a prototype data/ML editor based on a fork of VS Code for investor pitching.
- Adapted an Electron-based IDE codebase into a specialized editor concept for data science and ML workflows.

<https://arestov.github.io/linkkraft-notes/cv-media/atad-ml%20vscode%20fork.mov>

Career Break

Mar 2022 - Oct 2023

- Took a career break after Linkkraft's initial product phase.

Linkkraft

Founder / Principal Engineer | Nov 2021 - Feb 2022

Offline-first Electron browser and digital knowledge tool

- Built an offline-first Electron browser and digital knowledge tool for saving, organizing, and reopening web content through page snapshots and history-tree navigation.
- Implemented DOM snapshotting, resource capture, sanitization, caching, and deterministic offline replay so saved pages could be reopened close to the exact visual state seen by the user.
- Used Electron APIs and Chrome DevTools Protocol to capture resources, save styles, images, fonts, and responsive CSS, and reduce duplicate storage through hashing and reuse.
- Built canvas-based tree visualization for page-history navigation with selective redraw only when underlying data changed.
- Used DKT as the state/runtime layer with persistence, hydration, derived state, serialization, and selective loading/unloading for large local datasets.

Sidekick Browser

Principal Engineer / First Engineer | May 2019 - Oct 2021

Browser startup later acquired by Perplexity

- Joined as first engineer and bootstrapped an Electron-based browser product, helping ship an MVP for fundraising in roughly 2 months.
- Owned frontend/browser architecture across Electron, Chrome Extension, IPC, custom state runtime, browser automation, CI, packaging, signing, and performance work.
- Hired, managed, and mentored a team of 4 engineers while remaining hands-on.
- Replaced MobX-based state with a custom graph-based runtime later evolved into DKT; migrated core product logic from Electron into a Chrome Extension architecture.
- Improved startup experience with instant cached HTML restore while JavaScript loaded; improved search responsiveness by 2-3x through architecture, rendering, and data-flow optimization.
- Reduced browser-heavy memory usage through Chrome profiling, heap snapshots, Chromium-level debugging, object reuse, allocation reduction, and custom Babel/AST tooling.
- Designed unload/restore behavior for tabs to reduce memory usage while preserving user state; built CI for Electron packaging/signing on macOS and Windows.
- Built browser/Electron automation infrastructure with WebDriver and Puppeteer; contributed to a product ranked #1 Product of the Day and #1 Product of the Week on Product Hunt in 2020.

Earlier Experience

Frontend / Full-Stack Software Engineer | Aug 2008 - May 2019

- **Leroy Merlin:** JavaScript Lead Developer for market.leroymerlin.ru operations; moved browser testing from Java-based tooling to JavaScript/Puppeteer-style automation.
- **Motion Tailor:** built React/Redux frontend and Node/PostgreSQL backend pieces for a customized video production/rendering platform; created browser QA automation with WebDriver, headless Chrome, screenshots, Docker, and SemaphoreCI.
- **Trucker Path:** built React/Redux product interfaces for truckloads and shipping; worked on UX, A/B tests, monetization research, and backend automation in Clojure/PostgreSQL/MongoDB.
- **EuroChem:** built streaming DB-to-Excel export flows and incremental update logic for heavy business reports, improving memory usage and freshness.
- **Datalaboratory:** built Mosca JavaScript DOM/SVG rendering after D3-based exploration; delivered performance-sensitive interactive visualization in roughly 6 weeks.
- **Seesu Music:** founded and designed a browser-based music application combining multiple APIs, high-performance popup UI, request prioritization, custom rendering/state ideas, and Zoom UI; achieved 4.9/5 Chrome Store rating.

PUBLIC TALKS

- FrontendConf 2015 - Declarative Goal-Setting Programming Approach: https://youtube.com/watch?v=vjMqtRG_cV8
- HighLoad++ 2012 - Browser Comet Connections Multiplexing: <https://youtube.com/watch?v=5TEaxF9KP9A>

AWARDS / RECOGNITION

- Product Hunt 2020 - Sidekick Browser #1 Product of the Day / Week.
- Chrome Apps Contest 2010 - 1st place; Opera Widgets Challenge 2010 - 1st place; Tizen App Challenge 2014 - Honorable Mention.
- BitTorrent recognition - received a \$50 Amazon gift card for sharing browser media playback techniques later used in uTorrent monetization.